Dataset:

Name: Amazon Reviews 2023
Link: https://amazon-reviews-2023.github.io/
Download Link (Reviews):
https://huggingface.co/datasets/McAuley-Lab/Amazon-Reviews-2023/resolve/main/raw/review_categories/Electronics.jsonl?download=true
Download Link (Metadata):
https://huggingface.co/datasets/McAuley-Lab/Amazon-Reviews-2023/resolve/main/raw/meta_categories/meta_Electronics.jsonl?download=true

Dataset Profiling (100,000 sample):

Metadata:

Field	Observed types and counts	Missing	Empty containers
author	22 dict; 20 null; rest absent	99.98%	0
average_rating	100,000 float	0%	0
bought_together	100,000 null	100%	0
categories	100,000 list	0%	7,621 (7.621%)
description	100,000 list	0%	41,952 (41.952%)
details	100,000 dict	0%	299 (0.299%)
features	100,000 list	0%	22,430 (22.430%)
images	100,000 list	0%	5 (0.005%)
main_category	98,945 str; 1,055 null	1.06%	0
parent_asin	100,000 str	0%	0
price	41,823 float; 6 str; 58,171 null	58.17%	0
rating_number	100,000 int	0%	0
store	99,445 str; 555 null	0.56%	0
subtitle	33 str; 9 null; rest absent	99.97%	0
title	100,000 str	0%	0
videos	100,000 list	0%	58,860 (58.860%)

Reviews:

Field	Observed types and counts	Missing	Empty containers
asin	100,000 str	0%	0
helpful_vote	100,000 int	0%	0
images	100,000 list	0%	92,980 (92.980%)
parent_asin	100,000 str	0%	0
rating	100,000 float	0%	0

text	100,000 str	0.02%	0
timestamp	100,000 int	0%	0
title	100,000 str	0%	0
user_id	100,000 str	0%	0
verified_purchase	100,000 bool	0%	0

Schema Decisions:

Products:

Field	Data type	If missing or blank	Validation / handling
parent_asin	VARCHAR(20) **PRIMARY KEY**	Reject record	Must be unique
title	LONGTEXT	Store NULL	Must be text
average_rating	DECIMAL(3,2)	Store NULL	Between 1 to 5; otherwise NULL
rating_number	BIGINT	Store NULL	Nonnegative integer; otherwise NULL
price	DECIMAL(15,2)	Store NULL	Convert numeric strings; invalid, negative, or out of range values become NULL
main_category	TEXT	Store NULL	Must be text
store	TEXT	Store NULL	Must be text
categories	JSON	Store NULL	Must be a list; preserve []
description, features, details	JSON — separate columns	Store NULL	Description/features must be lists; details must be an object. And preserve empty containers
images, videos	Excluded	—	
author, subtitle, bought_together	Excluded	—	

Reviews:

Field	Data type	If missing or blank	Validation / handling
source_id, source_line	VARCHAR(100), BIGINT; **combined primary key**	NA	Combination must be unique
parent_asin	VARCHAR(20) NOT NULL **FOREIGN KEY**	Reject record	Import all first; delete unmatched reviews during cleanup.
user_id	VARCHAR(100)	Store NULL	Must be text
asin	VARCHAR(20)	Store NULL	Must be text
rating	DECIMAL(3,2)	Store NULL	Between 1 to 5; otherwise NULL
title	LONGTEXT	Store NULL	Must be text
review_text	LONGTEXT	Store NULL	Retain review even without text
reviewed_at	DATETIME(3)	Store NULL	Convert source timestamp milliseconds

			to UTC date/time; invalid values become NULL
helpful_vote	BIGINT	Store NULL	Nonnegative integer; otherwise NULL
verified_purchase	BOOLEAN	Store NULL	Only True or False
images	Excluded	—	